Predicting Pathway Left Shifts in Hospital Care using Predictive Process Monitoring

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Abstract. Hospitals worldwide are under pressure to use limited resources efficiently while improving healthcare outcomes. Patients typically follow a linear care pathway (CP), moving from admission to surgery, acute care, recovery, and discharge. A "left shift" occurs when a patient returns from less acute to more acute care, indicating deterioration and potentially a premature transfer decision or poor care planning. Such left shifts disrupt hospital processes and negatively affect both patient outcomes and costs. Predicting left shifts can lead to better outcomes and allow clinicians to plan and adjust treatment.

This paper focuses on left shifts to the intensive care unit (ICU) or surgical ward (SW) as markers of patient deterioration. Our review of the literature found that previous studies on predicting ICU or SW left shifts have largely overlooked process aspects. Process Mining (PM) provides deeper contextual insights that can improve prediction accuracy, and predictive process monitoring (PPM) leverages completed case traces to make predictions for ongoing cases.

We used the open-access MIMIC-IV dataset to evaluate the effectiveness of PPM in predicting ICU or SW left shifts, combining demographic, clinical, and process-related features. An event aggregation method was tested using machine learning (ML), deep learning (DL), and deep sequence (DS) models. Our results show that process-aware PPM significantly outperforms traditional prediction methods and demonstrates the potential to predict, plan for, or avoid left shifts, supporting smoother and more effective hospital care.

Keywords: Pathway Left Shift, Predictive Process Monitoring, Unplanned ICU Readmissions, Unplanned Surgery Ward Readmissions, Unplanned Reoperation, Clinical Pathway, Process Mining, Healthcare, Patient Flow Management, Event Log, EHR, MIMIC-IV, Machine Learning, Deep Learning, Deep Sequence.

1 Introduction

Healthcare is increasingly defined by the pressure to simultaneously achieve better patient outcomes and improved operational cost management. Hospitals are complex health systems responsible for delivering high-quality care and do this through standardised processes, procedures, technologies, and medications. Within a hospital, patients often follow a linear care pathway (CP) that is typically modelled as a process

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from left to right. For example, from planned or emergency admission, through surgery to acute care to recovery wards and through to dis-charge. We define a “left shift” as occurring when a patient returns back from less acute to more acute care, indicating a premature transfer, inadequate care planning or unresolved clinical issues [17]. Such left shifts to an intensive care unit (ICU) or surgical ward (SW) indicate a deteriorating patient, impacting patient outcomes and increasing healthcare costs [18].

A large U.S. cohort study across 156 ICUs reported that approximately 2% of patients were readmitted to an ICU within 48 hours after transfer to a ward, and 3.7% within 120 hours [17]. Another study showed unplanned ICU readmissions are associated with a mortality rate of 21 – 40%, compared to just 3 – 8% in patients not readmitted [18]. For unplanned re-operations (SW left shifts), one study found about 14% of patients with major head and neck surgery required unplanned reoperation during the same hospital stay [19]. Accurate prediction of these left shifts is vital for preventing poor outcomes, enabling proactive management, and optimising resource use [5]. Our review of existing literature on predicting ICU or SW left shifts found that most of the previous studies do not focus on prediction ICU or SW unplanned readmissions within same hospital stay and has largely overlooked process aspects. Our literature review found that pathway left shift prediction is limited to date, typically focusing on specific clinical specialties, narrow feature sets, and few predictive models. Some studies used machine learning (ML) [20][24], deep learning (DL) [21], deep sequence (DS) [23] and natural language processing (NLP) [22] to predict ICU or SW left shifts.

Process Mining (PM) of clinical event data offers deeper contextual insights, which can significantly increase the accuracy of predictions. In healthcare, PM has been shown to reduce patient waiting times, decrease costs, optimise resource use, and lower risks like morbidity and mortality [3]. Healthcare processes are modelled as event sequences capturing activities, timestamps, and related entities. Fig. 1 shows a simplified care pathway based on patient transfers, highlighting various pathway left shifts.

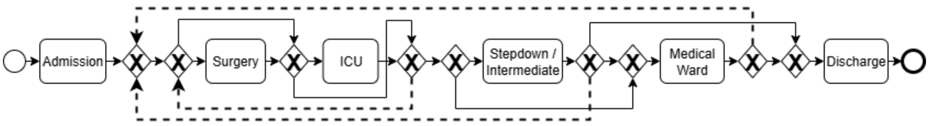

Fig. 1. High-level BPMN model of a generic hospital patients’ care pathway showing left shifts as dashed lines returning to an earlier step in the pathway.

Key PM techniques like process discovery and conformance checking are used to evaluate compliance with clinical guidelines. Prediction extends these benefits, as forecasting future events or outcomes is often more valuable than retrospective analysis, especially considering the high costs of delayed diagnosis or treatment [4]. Anticipating outcomes or next steps enables interventions that enhance care quality and efficiency [2]. Predictive Process Monitoring (PPM) is a process mining approach that leverages completed case traces to make predictions for ongoing cases. PPM enhances PM by allowing predictions for ongoing cases at different process stages [6]. These predictions can cover outcomes, next activities, activity sequences, execution time, or resource use. In healthcare, developing predictive capabilities is considered a major challenge and is identified as *Challenge 2: Discover Beyond Discovery* by the process mining for healthcare (PM4H) research community [1]. We identified only one previous study that has used PPM for predicting ICU left shifts [23].

In this study, we applied a range of established prediction models to forecast ICU and SW left shifts in a general hospital population using a PPM framework. Predictions are made at two points in the CP: (a) after the first ICU stay and (b) after the first SW stay. The remainder of this paper is organised as follows: Section 2 presents background and related work. Section 3 describes the PPM framework adopted. Section 4 details our implementation for pathway left shifts prediction, while Section 5 reports and discusses the results. The paper concludes with suggestions for future research.

2 Background

2.1 MIMIC-IV

Our study uses data from the Medical Information Mart for Intensive Care IV (MIMIC-IV), a well-known and openly available Electronic Healthcare Record (EHR) database [7]. We selected MIMIC-IV to implement PPM in accordance with Challenge 4: *Deal with Reality* from the PM4H community, which emphasises applying PM to real-world healthcare data [1]. Using MIMIC-IV also ensures our experiments can be reproduced by others. The database contains anonymised, detailed clinical data for ICU and ED admissions at Beth Israel Deaconess Medical Centre, Boston, USA, covering over 380,000 patients from 2008 to 2019. Its comprehensive event data makes it highly suitable for PM4H research [1].

2.2 Prediction in PM for Healthcare

Several methodologies have been introduced for managing PM projects. The widely adopted *PM2* model includes six stages and applies ML in the *Mining & Analysis* phase [8]. *ClearPath*, an extension of PM2, was created for CP discovery and integrates process simulation [9]. The *L* life-cycle* model adds prediction in its final *Operational Support* stage [10]. However, current frameworks lack specific guidance on how to implement predictive techniques within PM.

2.3 Predictive Process Monitoring

Like other PM methods, PPM depends on event logs that include at least a *case ID*, *activity*, and t*imestamp*. However, this basic data alone is usually insufficient for accurate prediction. To support predictive tasks, event logs should be enriched with *event attributes* (which change over time) and *case attributes* (which remain constant). These can be *categorical* (e.g., ethnicity), *numerical* (e.g., age), or *textual* (e.g., clinical notes). In healthcare, *static* attributes like age remain unchanged during admission, while *dynamic* ones such as blood pressure fluctuate due to repeated measurements.

A *trace* is a sequence of events for a single case, either complete or partial. In PPM, the partial sequence before a target event is called a *prefix*, which is used for prediction since PPM works on ongoing cases. Prefixes are converted into fixed-size *feature vectors* to label and train classifiers [11].

2.4 Pathway Left Shift Prediction in Healthcare

Through our literature review, we found few studies tried to predict pathway left shifts, focusing on predicting ICU or SW readmissions. Su et al. developed a high-performing Random Forest (RF) model (AUC = 0.889, accuracy = 84.2%) using proprietary rectal cancer surgery data to predict unplanned reoperations [20]. Inan et al. used 11,000 MIMIC-III ICU stays to predict readmissions via clinical features, with Artificial Neural Network (ANN) achieving AUROC = 0.874 but with class imbalance present [21]. Wu et al. used MIMIC-III data (36,232 ICU stay) to predict 30-day ICU readmissions by combining BERTopic-derived text features from discharge summaries with LSTM, achieving AUROC = 0.80 and demonstrating improved performance over models using structured data alone [22]. Rojas et al. developed a Gradient Boosting (GB) model using rich EHR data (24,885 ICU to ward transfer; validated on MIMIC-III) to predict unplanned ICU readmissions, achieving AUROC = 0.76 [24]. Finally, Chen et al. applied PPM to model ICU stays as event-log traces from MIMIC-IV and then predict ICU left shifts, demonstrating that early-stage prediction of unplanned ICU left shifts is feasible, achieving up to 65% accuracy using both clinical and process features [23].

3 Methodology

In this study, we adopted the Outcome-Oriented PPM (OOPPM) framework from [11] to predict left shifts. Developed based on a systematic review of OOPPM methods, this framework includes two main phases: an offline phase for training predictive models and an online phase for testing and real-time use (Fig. 2).

4 Predicting Left shifts Using PPM

This section details the implementation steps for each stage of the selected PPM framework. All experiments were conducted using Python 3.4 in Google Colaboratory (https://colab.research.google.com/).

4.1 Creating Extended Event Log

In this study, we used the MIMIC-IV dataset to construct an extended, labelled event log of hospital care pathways. While event log creation is not explicitly covered in existing PPM methodologies, careful development and enrichment of these logs with predictive features is a crucial step that requires special attention. In our adaptation to the OOPPM framework, we have added *'Create Extended Event Log'* as a separate step.

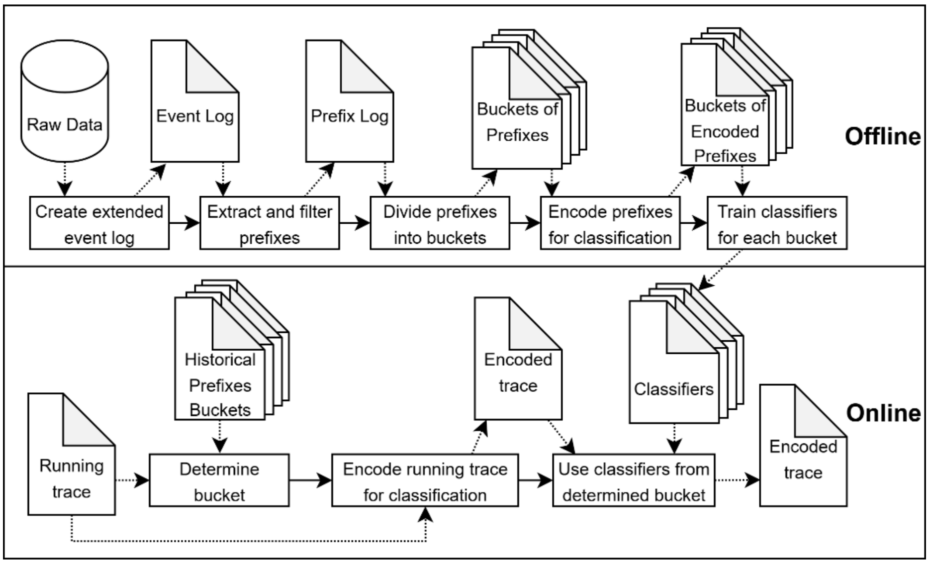

Fig. 2. OOPPM offline and online phases

Data Filtering. We modelled patients' clinical pathways using the *Transfers Table*, focusing on cases where patients were admitted to an ICU or SW at least once during their hospital stay. We have added a binary label to identify occurrence of the left shifts to ICU or SW for each hospital admission. Two event logs were created: the first includes transfer events for patients who went to an ICU ward including their ICU stay before transfer to another ward to predict the probability of their return. The second log contains the same for patients who went to an SW. Events with "Unknown" units were excluded, and the hospital admission number (*HADM_ID*) was used as the case identifier. The first log included 54,782 admissions and 234,294 events, and pathways ranging from 2 to 23 events - of these, 9,239 admissions had at least one left shift to ICU (16.8%). The second log had 61,312 admissions, 193,513 events, and clinical pathways ranging from 2 to 22 events - of these, 6,966 admissions had at least one left shift to SW (11.36%).

Extracting Data Features. Once the study cohort was defined, we organised the predictive features into demographic, clinical, and process-related categories.

Demographic Data. We selected *age*, *gender*, *insurance type*, *ethnicity*, and *marital status* as demographic features. The study included adults aged 18 to 91, with ages above 91 anonymised to 91 by MIMIC-IV [7]. To enhance prediction, we grouped ages into eight ten-year intervals, with the last group covering ages 88 and above. MIMIC-IV provides two gender categories, five marital status options, and three insurance types. The original 31 ethnicity categories were consolidated into eight main groups.

Clinical Data. Given the extensive clinical data in MIMIC-IV, we selected those clinical features used in established literature, prioritising those more likely to be clinically relevant for predicting left shifts to ICU or SW. We avoided using more complex data structures that require clinical domain knowledge to select and interpret (e.g., vital signs) [16]. Patient history was captured by counting previous *unplanned hospital readmissions* (UHRs) within 30 days of discharge following an approach we defined in our previous work [12]. We identified *ED admissions* and categorised the current *admission type*. *BMI* data was grouped into four categories: underweight, normal weight, overweight, and obese. To reflect patient health, we calculated the percentage of *abnormal lab results* per admission [15]. The number of *chronic diseases* was determined using AHRQ published list of codes [13].

Diagnoses in MIMIC-IV are recorded using ICD-9 and ICD-10. The dataset spans 2008–2019, with a transition from ICD-9 to ICD-10 during this period. Although ICD codes are generated when patient is discharged, we have used them to identify current diagnoses in the absence of other informative data. To address coding differences, we used 18 ICD-9 *disease categories* and calculated the number of diagnoses per category for each admission; the same approach was applied to ICD-10 records. This method minimised the need to encode thousands of individual ICD codes. Additionally, we computed the *Charlson Comorbidity Index* for each admission, as widely used measure of patient health status [14]. The *number of medications* administered to each patient was calculated, and patients with five or more medications flagged as *polypharmacy*.

Process Data. PPM should incorporate more than just control-flow events, since including additional process-related features can improve model learning and better leverage process-oriented data. In this study, we examined different aspects of the hospital admission process, with a focus on context, event frequency, and timing. For context, we identified 11 possible patient *admission locations*. For temporal features, we measured the *accumulated duration* after each transfer in hours to indicate how long the patient stayed in hospital. *ICU duration* and *SW duration* were calculated in hours for both event logs, up to the prediction event. The number of *ICU stays, SW stays*, and *total events* of each admission were also counted.

In total, we generated five demographic features, 27 clinical features, and seven process-related features, which were used in our classification models.

Extracting Events. Before building the extended event log for PPM, it was necessary to select events aligned with our prediction objectives. MIMIC-IV offers many event types, making this selection complex. While process discovery aims to capture all events, PPM benefits from a more targeted approach. Including too many low-value events can unnecessarily expand the feature space and hinder classification but omitting key events risks losing important process information. In this study, we focused on events from the *transfers* table, which captures patient movements around the hospital, providing a clear representation of each patient's hospital journey and supporting the patient-centred focus of PM4H [1].

Transfers events in MIMIC-IV are divided into 77 types: 38 for admissions to clinical units, 38 for transfers between units, and one for discharge. To maintain data quality and relevance, we excluded events labelled as "admitted into Unknown" or "transferred to Unknown." Based on clinical advice, we aggregated activities into 11 categories such as "Admission to ICU", "Admission to Surgery", "Admission to Stepdown / Intermediate" and "Admission to Medical Ward". We have followed the same for "transferred to" activities, while discharge events were not included in the event logs since we are predicting before that. Both original and aggregated event types were used independently to evaluate classifier performance and compare results.

The first event log for ICU left shifts prediction included 2,795 unique trace variants using non-aggregated transfer events, and 338 variants with aggregated events. For SW left shifts prediction, the event log had 2,115 variants with non-aggregated events and 356 variants when using aggregated transfers.

4.2 Implementation of PPM

Extracting and Filtering Prefixes. Three approaches exist in the literature for extracting and filtering prefixes: using a fixed number of events, fixing gaps, or defining a prefix by execution time [11]. However, due to the high heterogeneity of healthcare processes [1], and with the large sample size in our study, we needed to reduce the prefix log size. To address this, we focused solely on transfer events and applied aggregation for simplification. Cases with obvious data quality issues were excluded, and prediction points were set after patients were transferred from ICU or SW to another ward. Patients admitted for planned surgeries (Surgery → Discharge) were excluded to prevent data imbalance. Given the large number of trace variants, prefix filtering was not feasible, so we included all prefixes and acknowledge this may affect classifiers performance.

Divide Prefixes into Buckets. In this study, we used the "process states" or "decision points" bucketing approach by separating ICU and SW left shifts into distinct logs and with dedicated classifiers were trained and tested for each bucket. We also applied K-Means clustering with two clusters, selecting this configuration after evaluating various clustering methods. Given the high variability and heterogeneity of prefixes in our data, we did not use prefix-length-based bucketing.

Encode Prefixes for Classification. Classifier inputs require fixed-size feature vectors, so all bucketed traces of prefixes must be encoded into a consistent vector format. As event sequences grow, this becomes more challenging, and encoding must balance information loss with the need for generalisable methods.

Case and Events Attributes. Unique numerical attributes were directly included in the feature vectors and frequent numerical features were aggregated. Unique categorical attributes were one-hot encoded, while the frequency and duration of repeated categorical attributes were summed. All these attributes were converted to float. This encoding strategy, recommended in the literature, aims to improve classifier performance [11]. Labels representing the left shifts are encoded as 0 (negative) and 1 (positive).

Control-flow Aggregation. For ML models, bag-of-words (BoW) vector were used to encode activity sequence as features with N-gram range 1 to 3 to preserve some of the local process (i.e., activities order). For DL models, label encoder was used to assign each activity a unique number, converting each sequence of ordered activities into sequence of integer codes, then padded all integer sequences with zeros to the length of longest pathway so they can be fed into neural networks.

Train Classifiers for Each Bucket. The classification task in PPM is related to early sequence classification in ML [11]. Although standard metrics like *accuracy*, *precision*, *recall*, and *F1-score* are widely used, PPM also emphasizes *earliness* and *computation time*, which are critical for online predictions on ongoing cases. In this study, earliness was not assessed since our prediction points were fixed. However, computation time for training and testing was recorded to evaluate model efficiency. Additionally, Area Under the Receiver Operating Characteristic Curve (AUROC) and Area Under the Precision-Recall Curve (AUPRC) were calculated to provide a more comprehensive assessment.

We used Logistic Regression (LR) as a baseline for all experiments. The ML models trained and tested included GB, RF, and K-Nearest Neighbours (KNN). Given their strengths in sequence prediction [6], we also evaluated DL models: Multilayer Perceptrons (MLPs), Convolutional Neural Networks (CNN), and a CNN-LSTM hybrid. For DS models, we implemented Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Bidirectional LSTM (BiLSTM), and Gated Recurrent Units (GRU) and Bidirectional GRU (BiGRU). All models were trained with class weights applied, and hyperparameter tuning was performed. Cross validation with 10-folds was implemented, threshold for positive class and feature importance were checked.

We tested classifiers without process features, then added process features (except events), and finally included event features. We tested with and without clustering data and aggregating activities, to ensure completeness of the experiment.

Execution of Offline and Online Phases. To simulate the offline and online phases, we split the event logs into 80% for training and 20% for testing. A larger training set was chosen to support complex DL and DS models. Ideally, data should be split chronologically to mirror real-world scenarios, but this was not feasible due to the temporal anonymisation that was implemented when MIMIC-IV was created [7]. In the online phase, test data underwent bucketing and encoding before classification.

5 Results and Discussion

Table 1. Summary of classifiers results without process features.

Group	Classifier	Targeted Left shifts	Accuracy	AUROC	AUPRC	Precision	Recall	F1	Learning Time (S)	Prediction Time (S)
Base line	Logistic Regression (LR)	ICU	0.43	0.75	0.40	0.22	0.95	0.36	34.66	0.02
		SW	0.60	0.80	0.37	0.20	**0.83**	0.32	24.09	**0.02**
Machine Learning	Gradient Boosting (GB)	ICU	0.41	0.75	0.40	0.49	0.25	0.33	19.27	0.02
		SW	**0.89**	**0.82**	**0.46**	0.50	0.42	0.46	21.17	0.03
	Random Forests (RF)	ICU	0.83	0.74	0.37	0.49	0.22	0.30	9.19	0.34
		SW	**0.89**	0.80	0.44	**0.55**	0.37	0.44	9.11	0.33
	KNN	ICU	0.73	0.60	0.23	0.26	0.33	0.29	0.01	4.77
		SW	0.84	0.71	0.26	0.33	0.41	0.37	**0.01**	6.44
Deep Learning	MLPs	ICU	0.82	0.74	0.39	0.43	0.31	0.36	41.55	1.17
		SW	0.89	**0.82**	0.45	0.51	0.41	0.45	49.36	0.87
	CNN	ICU	0.82	0.78	0.41	0.45	0.31	0.38	42.68	1.33
		SW	0.86	**0.82**	**0.46**	0.42	0.52	**0.47**	44.06	1.34
	CNN + LSTM	ICU	0.81	0.75	0.41	0.42	0.37	0.39	42.85	0.72
		SW	0.87	**0.82**	**0.46**	0.43	0.51	0.46	46.28	0.82
Deep Sequence	RNN	ICU	0.82	0.74	0.39	0.45	0.27	0.39	48.95	1.02
		SW	0.88	**0.82**	0.44	0.47	0.43	0.45	55.61	1.33
	LSTM / BiLSTM	ICU	0.81	0.75	0.39	0.43	0.36	0.38	91.84	1.64
		SW	0.88	**0.82**	0.45	0.46	0.48	**0.47**	99.59	1.85
	GRU / BiGRU	ICU	0.82	0.74	0.39	0.44	0.31	0.36	92.88	1.79
		SW	0.88	**0.82**	**0.46**	0.46	0.49	**0.47**	88.87	3.21

Table 1 shows classifier results without process features for ICU and SW left shifts, while Table 2 shows results with process features included. Incorporating process features substantially improved predictions across all models. Without these features, the best ML model (RF) achieved 0.83 accuracy, and 0.30 F1 for ICU left shifts. For SW left shifts, RF and GB reached 0.89 accuracy but F1 remained below 0.47. DL and DS models showed similar trends, with ICU F1 not exceeding 0.39 and SW F1 below 0.47. LR had high recall (e.g., 0.95 for ICU) but low precision, reflecting its tendency to over-predict positives in imbalanced settings without informative features. Overall, the absence of process features caused large precision-recall imbalances and poor discrimination of true vs. false readmissions.

Table 2. Summary of classifiers results with process features.

Group	Classifier	Targeted Left shifts	Accuracy	AUROC	AUPRC	Precision	Recall	F1	Learning Time (S)	Prediction Time (S)
Base-line	Logistic Regression (LR)	ICU	0.84	0.95	0.79	0.52	0.95	0.69	11.92	0.01
		SW	**0.96**	**0.99**	0.86	0.72	**0.99**	0.84	7.79	**0.01**
Machine Learning	Gradient Boost-ing (GB)	ICU	0.94	0.98	0.91	0.79	0.87	0.83	14.75	0.04
		SW	**0.96**	**0.99**	**0.92**	0.76	0.98	0.85	13.07	0.03
	Random Forests (RF)	ICU	0.94	0.98	0.89	0.78	0.86	0.82	45.64	0.21
		SW	**0.96**	**0.99**	**0.92**	0.76	0.97	**0.86**	20.17	0.12
	KNN	ICU	0.89	0.89	0.72	0.65	0.78	0.71	0.01	42.15
		SW	0.93	0.94	0.73	0.64	0.84	0.73	**0.01**	29.31
Deep Learning	MLPs	ICU	0.92	0.97	0.88	0.73	0.87	0.79	37.64	0.67
		SW	0.95	**0.99**	0.89	0.73	0.97	0.83	52.45	0.71
	CNN	ICU	0.93	0.98	0.89	0.75	0.89	0.81	34.07	0.71
		SW	**0.96**	**0.99**	0.91	**0.77**	0.89	0.83	46.92	0.75
	CNN + LSTM	ICU	0.93	0.98	0.89	0.73	0.91	0.81	30.29	0.65
		SW	0.95	**0.99**	0.89	0.73	0.95	0.83	41.47	0.72
Deep Sequence	RNN	ICU	0.93	0.97	0.88	0.75	0.84	0.79	81.53	1.12
		SW	0.95	0.98	0.88	0.69	0.97	0.81	95.89	1.34
	LSTM / BiLSTM	ICU	0.93	0.98	0.89	0.76	0.88	0.82	161.45	1.85
		SW	**0.96**	**0.99**	0.91	0.76	0.96	0.85	177.41	1.87
	GRU / BiGRU	ICU	0.93	0.98	0.89	0.74	0.91	0.81	146.42	1.39
		SW	**0.96**	**0.99**	0.91	0.74	0.98	0.84	212.65	1.91

With process features, all classifiers showed performance gains. For ICU left shifts, RF accuracy rose from 0.83 to 0.94 and F1 from 0.30 to 0.82, while for SW left shifts, accuracy increased from 0.89 to 0.96 and F1 from 0.44 to 0.86. The best DS model (GRU) saw ICU F1 rise from 0.36 to 0.81 and SW F1 from 0.47 to 0.84. AUROC increased significantly, reaching up to 0.99, highlighting improved discriminatory power from richer temporal and process context. Precision and recall also became more balanced, indicating better model calibration and reliability.

Important process features for left shift prediction included *accumulated duration*, *ICU duration*, *surgery duration*, and *total events*; indicating that longer stays in acute units and more transfers suggest greater patient complexity and higher risk of a left shift in a patient pathway occurring. Including pathway activities improved F1 scores by up to 0.05 for ML models and 0.15 for DL models. Key clinical features were *abnormal lab tests*, *total medications*, and number of *respiratory diseases*. Clustering with two clusters had no impact on results. N-gram modelling showed no effect, likely due to high pathway heterogeneity, where activity presence mattered more than order. Activity aggregation slightly improved SW left shifts F1 by 0.01 but reduced ICU left shifts F1 by the same amount.

From a practical perspective, learning and prediction times are important for online PPM applications. Tree-based models like RF and GB offered fast predictions (<0.2s) and moderate training times, while DL and DS models required more time, especially with complex sequences. Still, the accuracy and reliability gains from process-aware models support their use in real-time settings, assuming computational resources and latency are managed appropriately.

6 Conclusion

Implementing PPM in this study was challenging due to the complexity and variability of healthcare processes. Unlike prior work focused on specific cohorts or long-term predictions, we targeted pathway left shift prediction for a general patient population, increasing heterogeneity. Tree-based ML models performed on par with advanced DL and DS models but were faster during the online phase, making them attractive for PPM in healthcare. However, process-intensive methods delivered higher predictive accuracy at the cost of increased processing time, limiting real-world applicability. Integrating PPM into Healthcare Information Systems (HISs) could address the need identified by the PM4H community to complement HISs with the process perspective [1].

Our approach in adding feature sets gradually, aggregating transfer events, clustering prefixes test, and assessing feature importance offers potential enhancements to current PPM practices. Nonetheless, effective feature engineering in PM4H requires more research and clearer guidelines, as healthcare processes are uniquely complex compared to other domains. Establishing standardised methods tailored to clinical data remains an ongoing challenge.

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